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Science and Medicine in Football

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/rsmf20

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To cite this article: Anne Hecksteden, Georges Pierre Schmartz, Yanni Egyptien, Karen aus der Fünten, Andreas Keller & Tim Meyer (2022): Forecasting football injuries by combining screening, monitoring and machine learning, Science and Medicine in Football, DOI: 10.1080/24733938.2022.2095006

To link to this article: https://doi.org/10.1080/24733938.2022.2095006

Accepted author version posted online: 27 Jun 2022.

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Publisher: Taylor & Francis & Informa UK Limited, trading as Taylor & Francis Group

Journal: *Science and Medicine in Football*

DOI: 10.1080/24733938.2022.2095006

Forecasting football injuries

by combining screening, monitoring and machine learning

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Abstract

Identifying players or circumstances associated with an increased risk of injury is fundamental for successful risk management in football. So far, time-constant and volatile risk factors are generally considered separately in either a screening (constant) or a monitoring (volatile) approach each resulting in a restricted set of explanatory variables. Consequently, improvements in predictive accuracy may be expected when screening and monitoring data are combined, especially when analysed with current machine learning (ML) techniques.

This trial was designed as a prospective observational cohort study aiming to forecast non-contact time-loss injuries in male professional football (soccer). Injuries were registered according to the Fuller consensus. Gradient boosting with ROSE upsampling within a leave-one-out cross-validation was used for data analysis. The hierarchical data structure was considered throughout. Different splits of the original dataset were used to probe the robustness of results.

Data of 88 players from 4 teams and 51 injuries could be analysed. The cross-validated performance of the gradient boosted model (ROC area under the curve 0.61) was promising and higher compared to models without integration of screening data. Importantly, holdout test set performance was similar (ROC area under the curve 0.62) indicating prospect of generalizability to new cases. However, the variation of predictive accuracy and feature importance with different splits of the original dataset reflects the relatively low number of events.

It is concluded that ML-based injury forecasting based on the integration of screening and monitoring data is promising. However, external prospective verification and continued model development are required.

Keywords

soccer, artificial intelligence, data integration, gradient boosting, prediction model

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Introduction

Competitive sports require athletes to approach their personal limits of performance and loading capacity. Not surprisingly, sport-related injuries are an issue in many sports, with football being a prominent example. Specifically, within a professional 25 player football squad around 50 injuries affecting the availability of the players for training and / or competition ("time-loss" injuries) may be expected per season (1-5). The negative consequences do not only concern the injured players health (and potentially career) but may also affect team performance (6) resulting in considerable financial losses for the clubs (7).

Consequently, understanding and preventing injuries is a priority issue. One important task in this context is to identify players or circumstances associated with an increased risk of injury. Plausibly, the focus will be on injuries that are relevant, do not announce themselves by warning symptoms but are plausibly predictable (at least in principle). This may translate to injuries that occur abruptly and affect the availability of a player (acute time-loss injuries). Regarding predictability, it has to be considered that for injuries due to physical contact with another player or object (contact injuries) irreducible uncertainty is plausibly high. In other words: Contact injuries happen because the player is in the wrong place and situation at the wrong time – with few if any risk factors that could be identified in advance.

So far, a considerable number of intrinsic and extrinsic risk factors for acute football injuries has been identified. The main extrinsic risk factor is the kind of exposure (e.g. match vs. training (1, 2)), but extrinsic risk factors also include framework conditions ranging from pitch surface (8-10), to match location (11), and current score (12). Intrinsic risk factors can be classified according to whether they change with time and if so, how fast. While some characteristics are more or less immutable in adults (e.g. body size (13, 14) or playing position (12, 14)) others change swiftly such as training load features (15-22)) and recovery needs (18, 23). Many other potential risk factors are modifiable in principle but major changes occur only slowly and require targeted interventions (e.g. body composition (13), different aspects of physical fitness (20, 24-31), and neuromuscular qualities (32)). This diversity is reflected in two approaches to estimating injury risk: (a) Within the screening approach players are periodically tested for relatively time-stable characteristics (e.g. anthropometrics (13, 14), different aspects of physical fitness (20, 24-31), and neuromuscular qualities (32)). Subsequently, injuries are recorded over a prolonged period of time, typically the following season. (b) Within the monitoring approach volatile factors such as training load and / or responses to it (e.g. strain, recovery needs and subjective wellbeing (12, 15-17, 28, 33)) are continuously recorded and injury risk is estimated for individual

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players and time points. Known basic player information (e.g. age (13, 14), playing position (12, 14)), and the history of previous injuries (11, 13, 34) may be considered in either approach.

Considering the panel of risk factors as a whole, the highly plausible scenario emerges that well-conditioned and well-recovered young players who so far did not get injured and who exercise under familiar conditions in the absence of competitive pressure have a low risk of injury. However, when trying to advance towards the quantitative estimation of injury risk needed to inform decisions in sports practice, predictive accuracy turns out to be impracticably low (35). For individual risk factors this is expectable based on their multitude and (at least partial) independence. Linear combinations of risk factors ("scores", "multivariate Cox regression models") with better performance have been developed. However, from the small number of proposed models (13, 18, 36) none has been scientifically confirmed outside the original setting.

Consequently, calls for advancing beyond linear models with additive effects of individual risk factors towards a non-linear "web of determinants", have been brought up for more than 20 years (34, 37-40) and in the meantime causal concepts for sport injuries have also steadily evolved in this direction (38-42). The lagging adoption in empirical work is plausibly due to 3 overlapping restrictions: (a) the specific framework conditions in high-performance environments (e.g. the mismatch between the small number of elite players and the requirements of machine learning (43)), (b) the challenges of longitudinal data collection with a comprehensive panel of risk factors (plus the recording of injuries on top), and (c) the availability of appropriate data analysis techniques. Importantly, algorithms aiming at practical forecasting accuracy must be able to cope with a scaring panel of challenges: Specifically, the already difficult combination of a high-dimensional, non-linear "web of determinants" with a small numbers of cases (43), is complicated by the pronounced imbalance between days with and without injury occurrence ("needle in a haystack"), and the hierarchical data structure (nesting of timepoints within players). However, recent technological advancements with respect to data collection and analysis (e.g. smartphone apps and advanced machine learning techniques) may today enable addressing these challenges.

Of note, implementing the "web of determinants" concept is not only a question of fancy data analysis but requires complementary data collection. Specifically, the explanatory variables available in the dataset have to cover all parts of the "web of determinants" illustrated in the conceptual figure below (fig. 1). Though, in the small number of published papers using machine learning to predict injury risk in football (15, 44-47), other sports (46, 48) or demanding physical exercise more general (49) the datasets remained confined to either the

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“screening” (44, 45, 49) or “monitoring” (15, 47, 48) approach. While the cross-validated performance metrics (e.g. AUC) indicate at least moderate performance in most cases, holdout test set performance is rarely provided and convincing prospective verification is missing (or failed when attempted (45)).

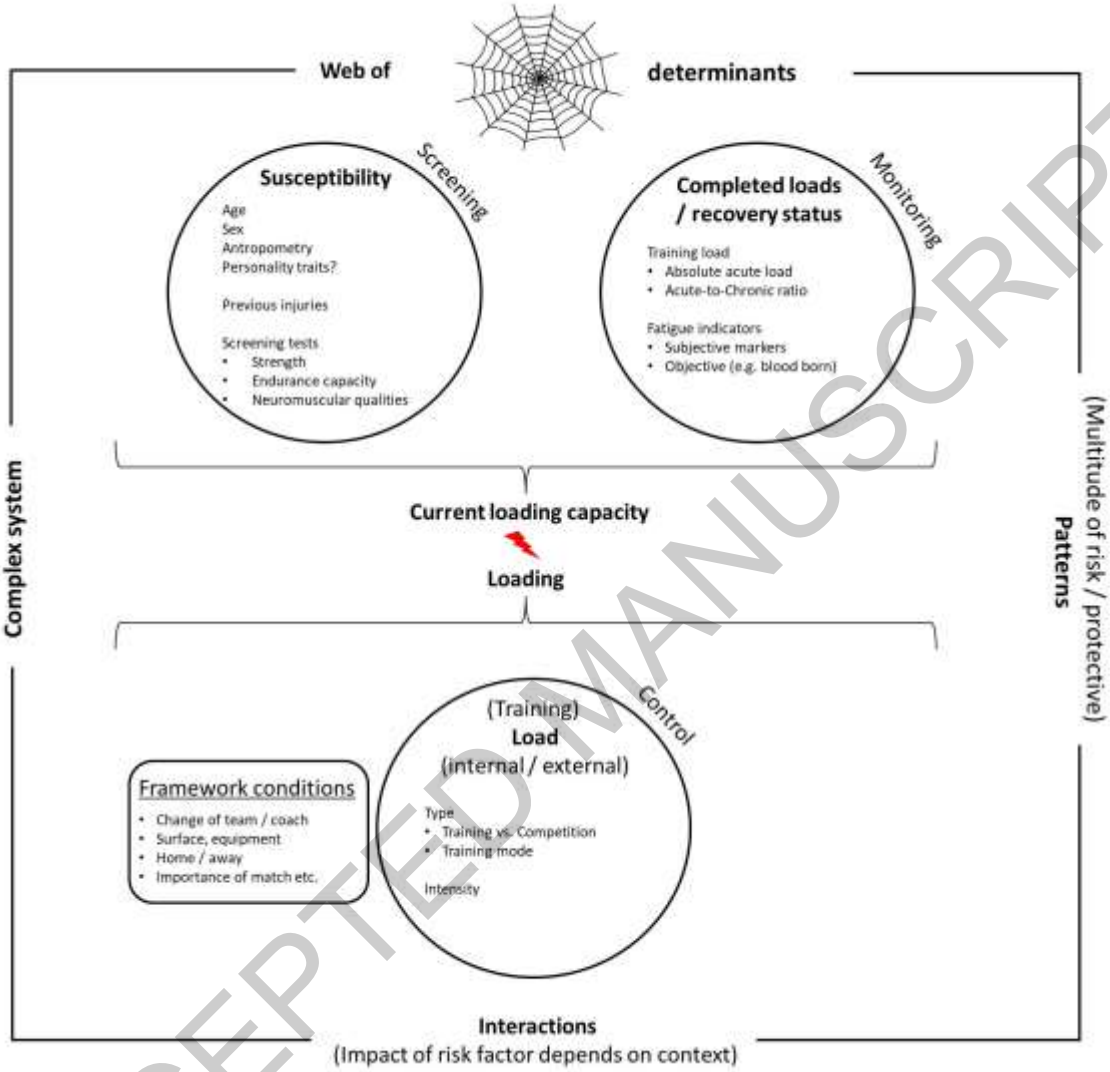

Figure 1: Conceptual figure for the “web of determinants” of football injuries

Based on these considerations, one may expect improvements in predictive accuracy when “screening” and “monitoring” data are integrated and analysed with state-of-the-art analytical techniques able to leverage the “web of determinants” concept while respecting methodological constraints (small number of cases, imbalance, hierarchical data structure). Plausibly, the “screening” data set the individual players level of risk while volatile “monitoring” data and extrinsic factors cause short-term fluctuations around this anchor.

This present work aimed to train and validate an injury prediction algorithm based on the above rationale in male adult professional football (soccer) players. Considering the limited size of the dataset, specifically regarding the number of criterion injuries, care has been

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taken to avoid overfitting and probe the robustness and potential generalizability of results. Despite these efforts, it is understood that the results (including the trained model) will be preliminary, and prospective validation will be indispensable before productive operation and transfer into sports practice may be envisaged. Model performance metrics have been chosen to allow comparison to common standards and previous work as well as an appraisal of potential practical utility.

Methods

Overall design

This trial was designed as a prospective observational cohort study. The aim was to develop a prediction model for non-contact time-loss injuries in male professional football based on the "web of determinants" concept illustrated in figure 1. Injury risk is to be predicted on the level of player and day. The study was conducted during the 2019/2020 football season. In total 112 professional male adult football (soccer) players from 4 teams of the German 4th and 3rd leagues have participated of whom 88 could be included in the analysis. The study was conducted in accordance with the Declaration of Helsinki and approved by the responsible ethics committee (Ärztekammer des Saarlandes, ID 230/19). All participants provided written informed consent.

While a comprehensive panel of potential explanatory variables has been collected, the expected number of criterion injuries (size of the minority class) implies a limited set of explanatory variables for the primary analysis. To avoid overfitting, a set of 11 explanatory variables was specified a priori (before the start of data collection), the period after return-to-play was amended before the start of data analysis. An overview is provided in table 1. Gradient boosting stood out as a suitable approach to data analysis because it combines conceptual consistency with the web-of-determinants concept and a fitting empirical compromise between model complexity and potential to learn under limited amounts of data. (Other plausible methods have been implemented for comparison as ancillary analysis 4. Please compare supplementary Markdown file Ancillary_4 and its knit html.) Upsampling was implemented to account for the imbalance between days with and without injury occurrence thereby improving discrimination (e.g. ROC-AUC). Upsampling consists in the generation of a balanced synthetic training-set based on the training-split of the original dataset. Considering the pronounced imbalance and the limited absolute number of injuries, synthetic datapoints were constructed (ROSE method) instead of simply oversampling the minority class. Details are provided below and in the accompanying code. Of note: The benefits and drawbacks of upsampling depend on the framework conditions of the usecase. Ancillary analysis 3 probes the issue for our specific dataset. (Please compare supplementary

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Markdown file Ancillary_3 and its knit html (cp. link to repository below) and the respective paragraph of the discussion section)

Data collection

Within this study, 5 main chunks of data were collected. The following section provides a concise overview. A detailed description of procedures is provided as supplementary file.

(a) Basic player information After providing written informed consent, participating players were asked to complete a custom questionnaire. Besides basic personal information this questionnaire also included the history of previous injuries, dominant leg, previous transfers and other football-specific details. Personal communication with the player and / or staff was used to clarify ambiguities.

(b) Periodic screening tests were conducted during the pre-season and during the winter break, respectively. In accordance with common practice in professional football screening included tests for different aspects of physical performance (including a stepwise incremental running test to determine endurance capacity (29, 30) and a 30 m linear sprint test), a test for movement quality (football injury movement screen (50)) and anthropometrics including body composition (51). Anthropometric measurements were carried out according to international standards for anthropometric assessment (52).

(c) Daily monitoring included internal training load (session RPE method (53)) and subjective recovery status (short recovery and stress scale (SRSS) (22)). Monitoring data were collected via a custom smartphone application developed during a previous project (54, 55). Exposure time was also gathered from those entries.

(d) Type of exercise exposure (match vs. training) and framework conditions (e.g. home vs. away and pitch surface) were recorded from online sources and direct communication.

(e) Injury recording and classification. To ensure all time-loss injuries are identified several approaches were combined (signaling by the player via the above smartphone app, regular contact with the responsible member of the medical staff within the club and the team physician). After identification, injuries were worked up in more detail and classified according to the Fuller consensus statement (56).

Data analysis

All analyses were conducted in R (version 3.6.3 (57)) The main specific packages used were tidyverse (58, 59), runner (60), lme4 (61), and quantable (62) for data preparation and feature engineering, ROSE (63) for upsampling (creating a balanced synthetic training set),

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and rpart (64) and caret (65) for fitting the gradient boosted model with repeated leave-one-out cross-validation.

The main analysis is tuned to mitigate and probe overfitting by considering the relatively low number of injuries, the marked imbalance in the dataset as well as the hierarchical data structure (nesting of time-points within players) (cp. below and figure 2). Specifically, this concerns a priori feature selection, estimation of holdout test-set performance, as well as a stratified train-test split and cross-validation on the level of players (not datapoints) to avoid leaking of information between folds. Upsampling was integrated into cross-validation to further reduce potential information leakage between folds and overestimation of cross-validation performance.

Ancillari analyses are implemented (1) to probe the robustness of results by varying the split of the original dataset in a round-robin manner, (2) to demonstrate the risk of spuriously high cross-validation performance when the nesting of time-points within players is not considered during cross-validation (3) to demonstrate the decrease in model performance when upsampling is omited. An overview is provided in figure 2. Ancillary analysis 4 demonstrates results for other plausible methods (please compare supplementary Markdown file Ancillary_4).

Densely commented, reproducible statistical code as well as the knit R Markdown files for feature engineering and data analysis, respectively, are made available in the following public repository (doi: https://zenodo.org/badge/latestdoi/493978425) . Providing raw data is not possible due to the challenges of effective data anonymization at this level of play and lack of consent.

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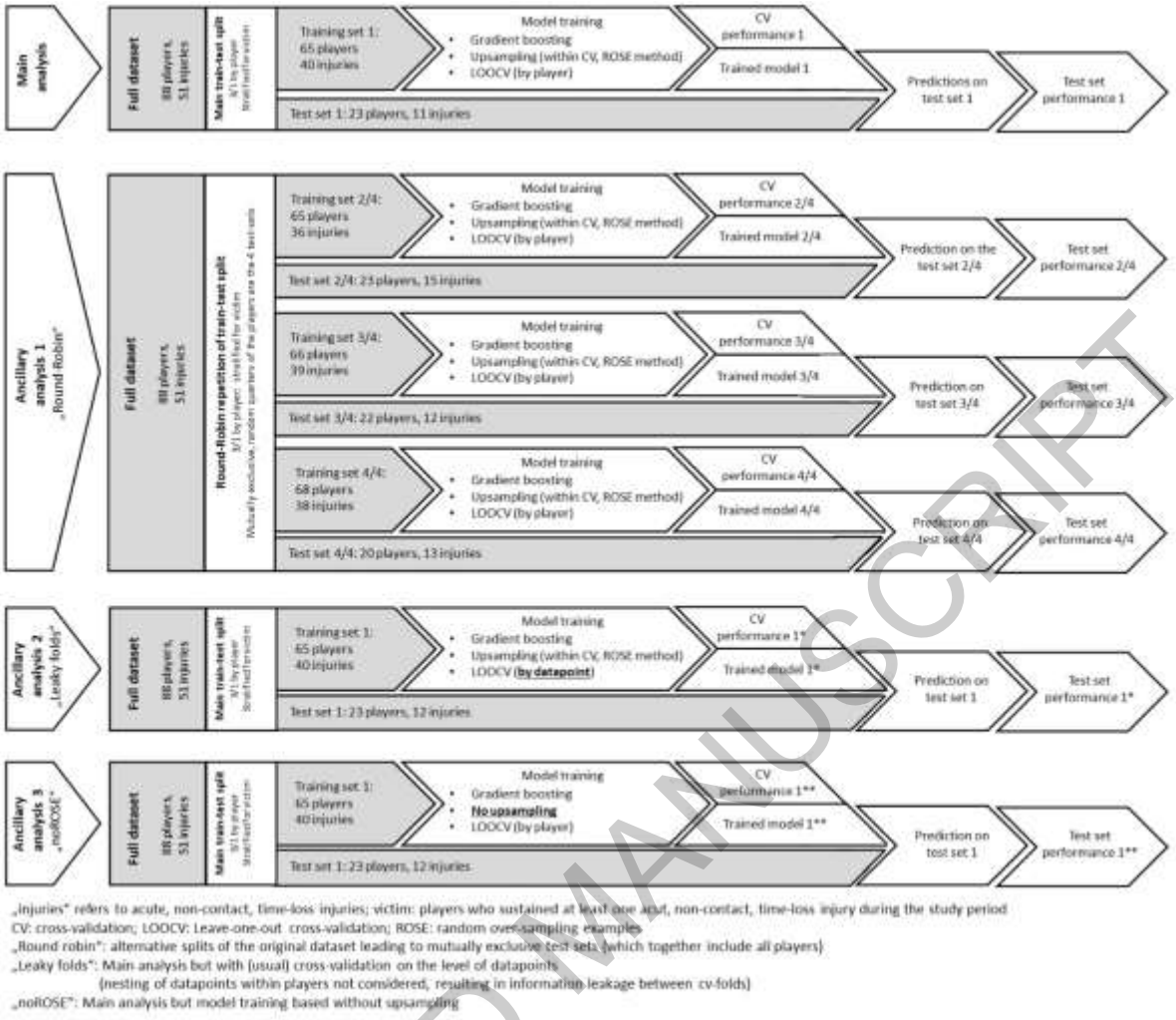

Figure 2: Data analysis and flow of information

<u>Data preparation and feature engineering</u>

Time frame to be included

Due to the Covid-19 pandemic, the 2019-2020 football season was interrupted on March 17th 2020. Although data collection continued and competitive play resumed in April, datapoints from after March 16th 2020 were excluded from the primary analysis because it was felt that conditions for injuries had changed too much. The effective study period was June 1st 2019 to March 16th 2020.

Preparatory calculations and classifications (feature engineering)

Considering the limited amount of data and the wealth of previous literature on individual risk factors for football injuries (cp. above), the set of explanatory characteristics has been specified a priori in a hypothesis driven manner (table 1). The following table provides a concise overview including rules for eventual imputation of missing values, adjustment for confounders and scaling. Please compare the commented statistical code for reproducible detail.

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Table 1: Criterion and explanatory features	
Criterion	
Day of injury	Day with occurrence of a non-contact, time-loss injury vs. day with full availability of the player *Days with existing injury, rehab training or otherwise limited availability for training and matchplay were excluded.*
Explanatory features	
Basic player information	
Age	Age [years] *Raw values are used for analysis*
Playing position	4 categories (Goalkeeper, Defender, Midfielder, Striker) *Original categories are used for analysis*
Injury history	Days lost to injury previous to the current day *Days lost during the study period were incrementally added to the baseline (days lost to injury previous to the study start).* *To account for the accumulation of exposure over time, we adjusted for age using the residuals of a linear model.*
Screening	*Imputation rule for all screening tests: In case of missing values for one of the two screening tests, the value from the other test was used throughout. The 3 players who missed at least one subtest on both occassions were excluded.*
Endurance capacity	Individual anaerobic threshold [km/h] (derived from an incremental running test to exhaustion (29, 30))
Sprinting speed	Time for a 30 m linear sprint [s] (light-gate timing, mean of 2 trials)
Body composition	Body fat [%] (10-point caliper method (51))
SIMS Score	Overall score in the Football Injury Movement Screen (50) [ ]
SIMS Pain	Pain in at least on e subtest of the Football Injury Movement Screen (50)
Monitoring	*Rules for imputation and scaling of monitoring parameters: Imputation of missing values was based on within-team central tendency for that day. To get an indication of how high or low training load is for the individual player, a robust scaling with median and interquartile range was applied on the individual level.*
Training load	Trailing, linearly weighted 7-day average of internal training load (sessionRPE method (53)) [ ] *In case of a missing entry for a specific day, the team average for that day was imputed. This step, as well as the following calculation of the linearly weighted, trailing 7-day average were executed by the custom software used for data collection.*
Subjective recovery	Dimension “overall stress” of the “short scale of recovery and stress for

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	sports" (22) *In case of a missing entry for a specific day, the team median for that day was imputed. In case this was not successful (whole team missed out that day) the interpolation between the team medians the day before and after was used.*
Exposure	
Type of load	Competitive match vs. training (or friendly match)
Stress imposed	Team average of sessionRPE *Robust scaling was applied globally.*
Special vulnerability	
Period following RTP	Days within a 4-week period after return-to-play (full availability for training and matches)

Modell fitting and evaluation

Train-test split

The curated dataset was randomly split on the player level into a training (3 quarters) and a holdout test set (1 quarter). To ensure equal proportions of the minority class between sets, the split was stratified between players who sustained at least one criterion injury within the study period ("victims") and those who did not.

Upsampling

As expected, the dataset was severely unbalanced with a marked underrepresentation of days with injury occurrence. Therefore balanced synthetic training data are generated during cross-validation by upsampling only the fold used for model evaluation. The ROSE method (random over-sampling examples, (63)) as implemented in the respective R-package (63) was used.

Model fitting and cross-validation

A gradient boosted model was fit using repeated leave-one-out cross-validation on the player level (repeatedly fitting the model with all but one player and assessing performance based on prediction for the left-out player until all players have been the left-out one). The 10 repetitions were used to tune interaction depth and number of trees. ROC area-under-the-curve (ROC-AUC) was used as main metric because of standard character and appropriateness for unbalanced data. As an ancillary metric, Brier scores (66, 67) are

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provided in the supporting Markdown documents to evaluate the model from the perspective of injury-forecasting. Brier scores are used to assess the accuracy of probability predictions for dichotomous outcomes. They are calculated as the squared difference of a probability prediction (between 0 and 1) from the true outcome (0 or 1) (66, 67). Confidence intervals are derived by bootstrapping.

Assessing holdout test set performance

The trained model was used to predict the outcome in the holdout test set (as well as in the original, non-upsampled training set, cp. Markdown file of main analysis). The same model-performance metrics as in cross-validation were applied. Moreover, to provide an intuitive assessment of potential contribution to decision making in sports practice, the relation between the proportions of time-points identified as "high-risk" and identified injuries, respectively, is visualized ("proportion needed to prevent").

Ancillary analysis 1: Round-robin

To estimate the robustness of results and get comparable ("test set") predictions for all player, the above pipeline was repeated 4 times in a round-robin manner. Specifically, before the train-test split of every round players who already had been included previously in a test set were withheld and later added to the training set. The train-test split ratio was adjusted accordingly to obtain equally sized test sets. The further steps were unchanged as compared to the main analysis.

Ancillary analysis 2: "Leaky-folds"

When cross-validation folds are not strictly separated, cross-validation performance is inflated and not representative of true test set performance. In the specific case of this analysis, two potential sources of information leakage are noteworthy: (a) The hierarchical data structure. In monitoring datasets, timepoints are nested within players. If this hierarchical data structure is not considered, datapoints from the same player will be included in various folds and hence may be used for training as well as testing the same model. Consequently, in the main analysis cross-validation folds are created on the level of

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players (not datapoints). (b) The upsampling procedure. When the whole training set is upsampled before cross-validation, resamples or descendants of the same case will be included in various folds and hence used for training as well as testing the same model. Therefore, in the main analysis upsampling is integrated in to cross-validation and cross-validation performance is based on non-upsampled data points.

In previous work these aspects are (as far as evaluable) generally not adressed. Moreover, holdout test-set performance is rarely reported, making it difficult to identify potentially inflated cross-validation performance. To demonstrate the point, ancillary analysis 2 repeats the main analysis with the only difference of not considering one or both of the two potential sources of information leakage.

Ancillary analysis 3: “noROSE”

Marked underrepresentation of the minority class (here: days with injury occurrence) negatively affects the performance of the trained model. Upsampling methods (e.g. ROSE) aim to mitigate the negative effects of class imbalance by generating balanced (or less imbalanced) synthetic training data. However, it has recently been argued that at least for some analytical methods (logistic regression (68)) upsampling may not only negatively affect calibration but also fail to improve discrimination as e.g. measured by ROC AUC. Therefore, ancillary analysis 3 implements the main analysis but without upsampling.

Comparator models

To probe the value of integrating screening and monitoring data, comparator models with only the monitoring data as well as with monitoring data and basic player information were trained. A comparator model with screening data only was not appropriate due to the much longer timeframe of predictions (usually a full season). Apart from subsetting features in the training set before upsampling, the modell fitting and evaluation process was unchanged. Of note, the ancillary analyses were not performed for the comparator models.

Results

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Dataset and participating players

17 218 datapoints ("player-days") from 88 individual players (12 goalkeepers, 22 defenders, 37 midfielders, 17 strikers) were included in the main analysis. Baseline player characteristics were representative for the level of play (means +- sd; age 24.6 +- 3.5 years, height 1,83 +- 0.07 m, weight 79 +- 7 kg, body fat 12.4 +- 3.0 %, individual anaerobic threshold 14.3 +- 0.8 km/h, 30 m linear sprint time 4.1 +- 0.2 s). The median number of days lost due to previous injuries was 90 days with a wide interquartile range of 12 to 223 days.

Epidemiological indicators

For the players included in the final analysis a total of 102 injuries has been recorded during the study period. Of those, 93 were time-loss injuries and 51 were acute, non-contact, time-loss injuries (criterion). Overall incidence per 1000 hours of exercise exposure was 3.9 for time-loss injuries (training: 2.7/1000h, competitive matches: 9.5/1000 h) and 2.2 for criterion injuries (training: 1.8/1000 h, competitive matches: 3.7/1000 h).

Injury prediction

Model evaluation

ROC curves are presented in figure 3. Cross-validation as well as holdout test set ROC AUCs for the main analysis as well as for ancillary analyses 1 to 3 are summarized in table 2. The violin plots in figure 4 illustrate the upward shift in the distribution of predicted injury probabilities for the days with injury occurrence (discrimination). Thereby, figure 4 intuitively confirms the findings in table 2 and figure 3. Note the varying degree of overlap in the distributions between modalities.

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Table 2: Model evaluation results

	ROC AUC	
	Cross-validation	Test set performance
Main analysis	0.61 [0.60 | 0.62]	0.62 [0.49 | 0.76]
Ancillary analysis 1: Round-robin		
Round 2/4	0.57 [0.56 | 0.58]	0.62 [0.49 | 0.75]
Round 3/4	0.60 [0.59 | 0.61]	0.69 [0.53 | 0.86]
Round 4/4	0.56 [0.55 | 0.57]	0.68 [0.53 | 0.84]
Ancillary analysis 2: Leaky folds		
Leak 1: No fold index	0.57 [0.57 | 0.58]	0.59 [0.42 | 0.75]
Leak 2: Global upsampling	0.66 [0.65 | 0.66]	0.64 [0.49 | 0.78]
Leaks 1 & 2	0.88 [0.88 | 0.88]	0.57 [0.37 | 0.77]
Ancillary analysis 3: noROSE		
noROSE	0.52 [0.51 | 0.53]	0.48 [0.32 | 0.64]

ROC AUC: Estimate [95% confidence interval (bootstrap)]

ROC: receiver operating characteristic curve; AUC: area under the curve

Round robin: Same procedure as main analysis but with different splits of the original dataset (mutually exclusive, random test sets).

Leaky folds: Same as main analysis but with two potential sources of information leakage: (1) No fold-index: cross-validation on the level of datapoints instead of players (that is: without considering the nesting of timepoints within players). (2) Global upsampling: Upsampling the whole training set before model training (instead of only the training part of each cross-validation fold).

noROSE: Same as main analysis but without upsampling (model fitting with imbalanced training data)

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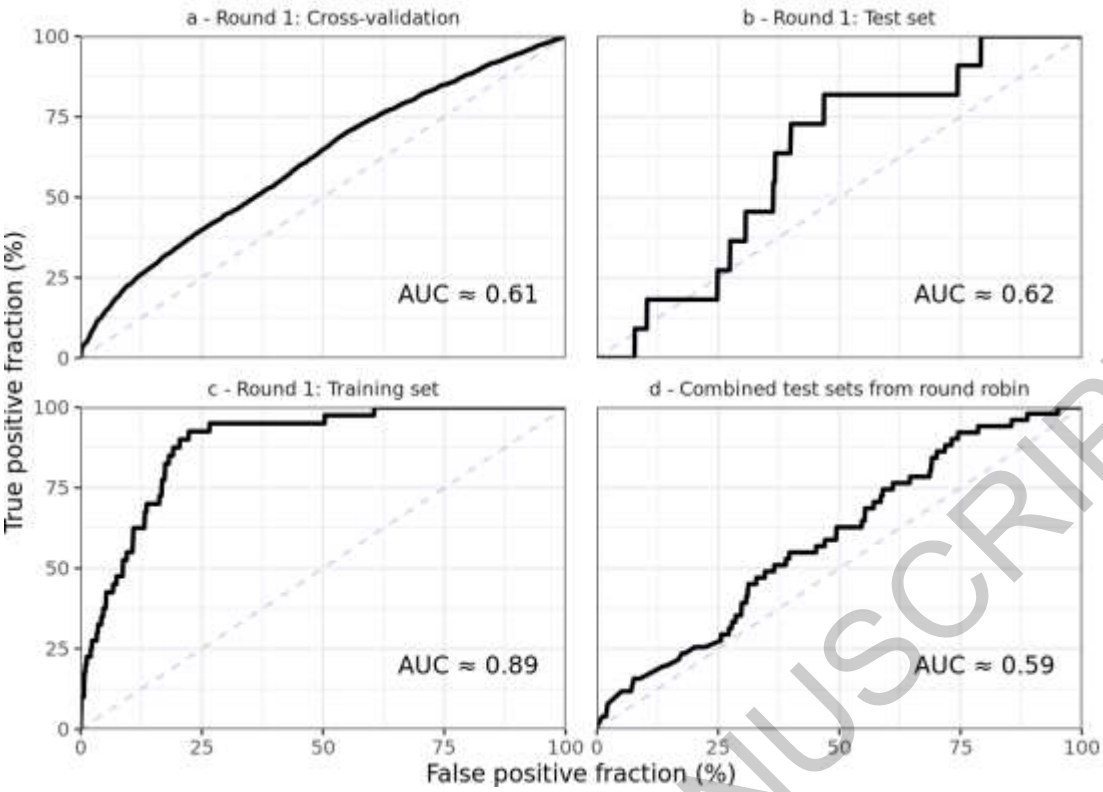

Figure 3 – ROC curves

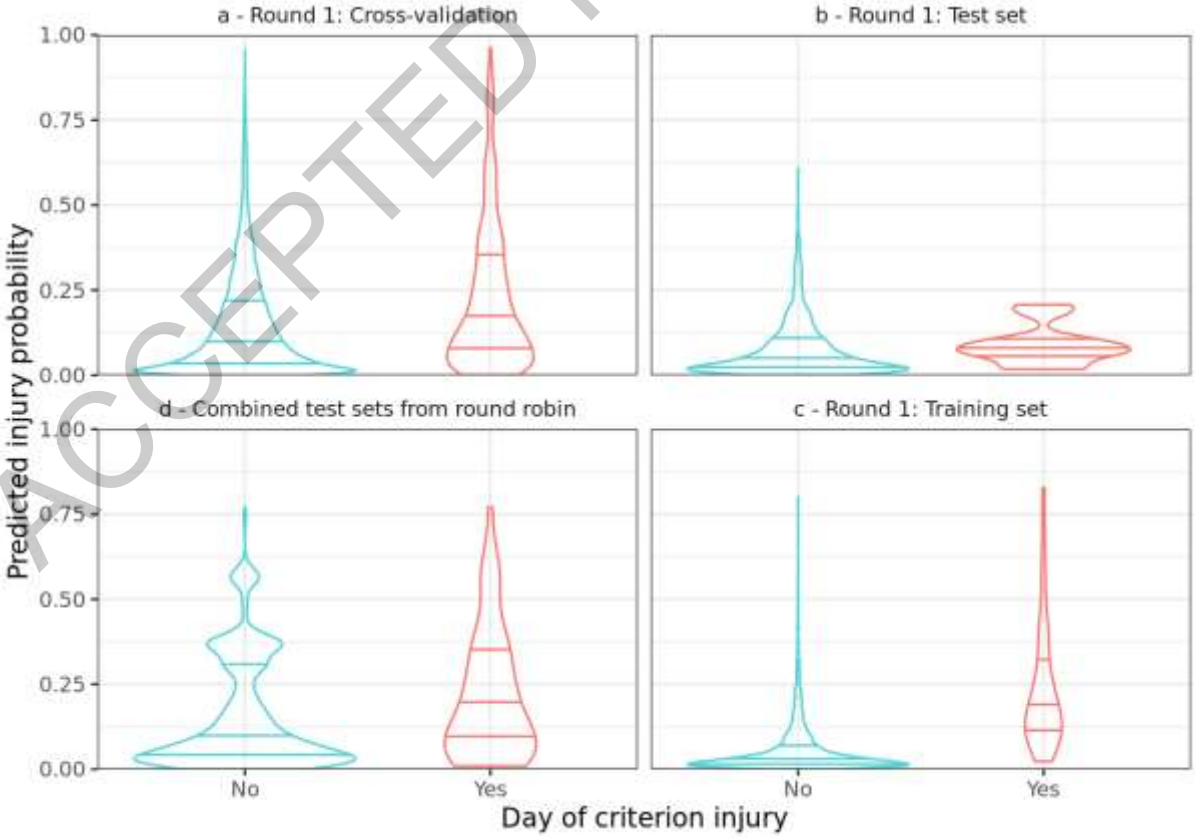

Figure 4 – Violin plots of predicted injury probabilities

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Potential relevance for decision-making in sports practice

Figure 5 visualizes the relation between the respective proportions of time points flagged as high-risk (that is: which would potentially be subject to consequences / preventive measures) and identified (that is potentially prevented) injuries. For predictions on the full dataset based on the main model, almost 1/2 of criterion injuries concentrate in 1/8 of time-points with the highest predicted injury probability. No injuries occurred in the ¼ of timepoints with the lowest probability estimates. For pooled test-set predictions, the relation is ~ 1/2 of criterion injuries in 1/3 of time points.

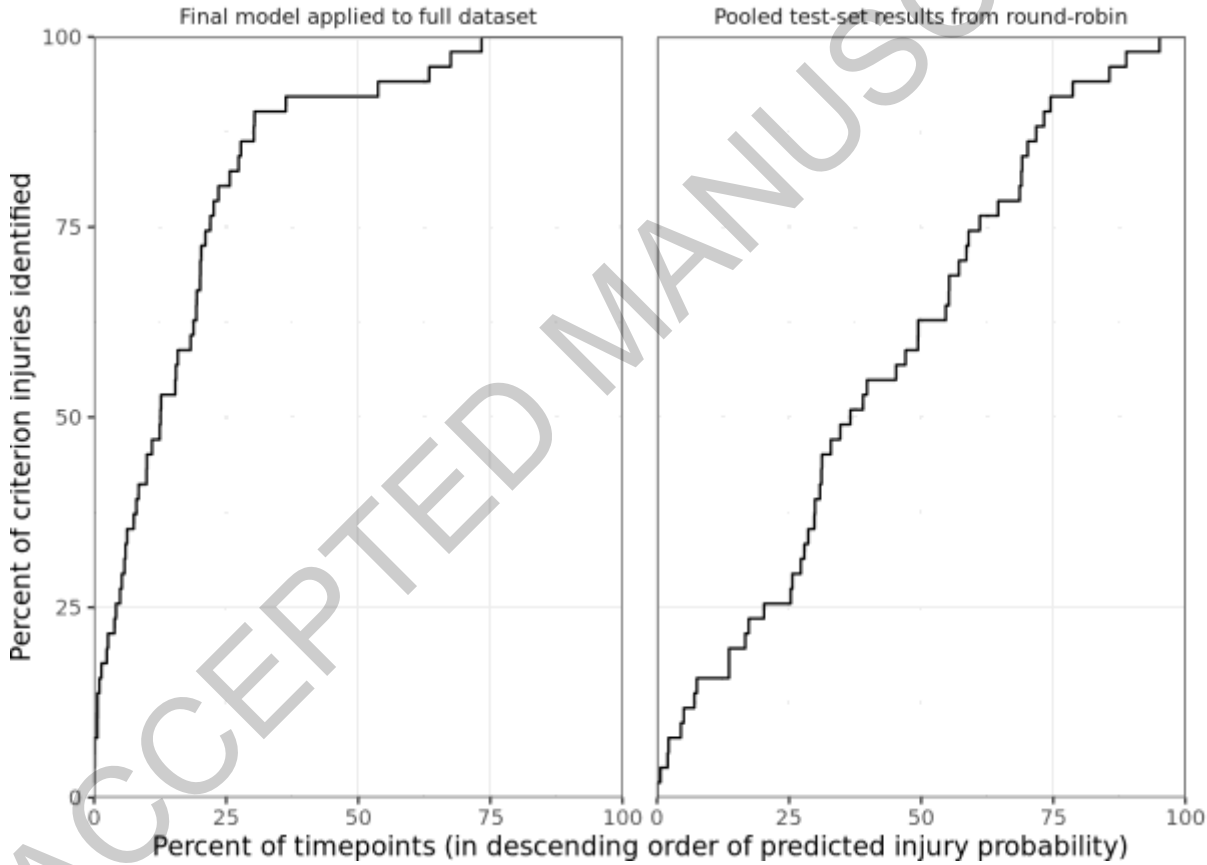

Figure 5 - Proportions of time-points identified as “high-risk” and identified injuries, respectively

Distribution and time course of predicted injury probabilities

Figure 6 visualizes the time course of probability predictions (pooled test set results) together with the timepoint of eventual criterion injuries. The subpanels represent the individual

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players during there respective periods of involvement in the study. Note the different personal levels of predicted injury risk (set be the time-invariant features) and the short term variations around that level. For a more detailed view please refer to the supplementary Markdown file.

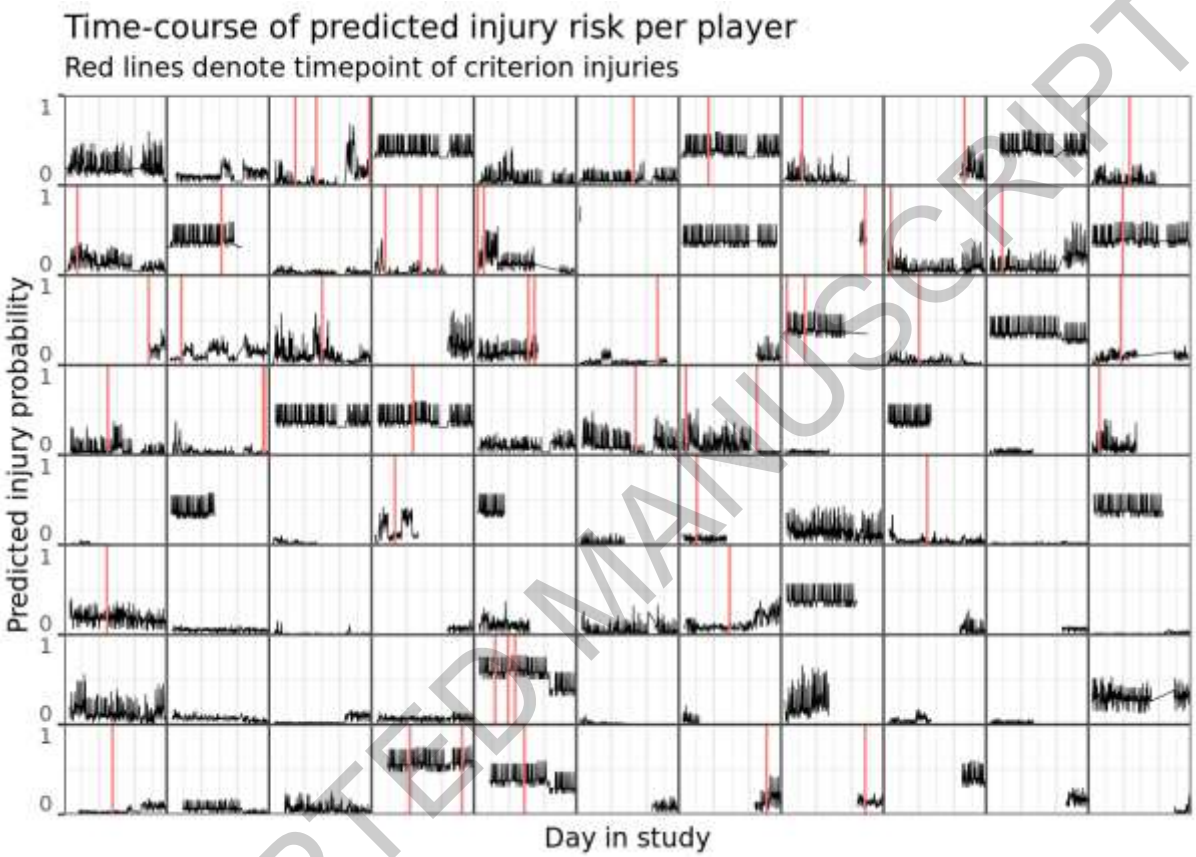

Figure 6 – Timecourse of probability predictions

<u>Feature importance</u>

Feature importance across the main model an the round-robin repetitions in illustrated in figure 7. While the overall variability is considerable, some features consistently seem to be of at least moderate importance.

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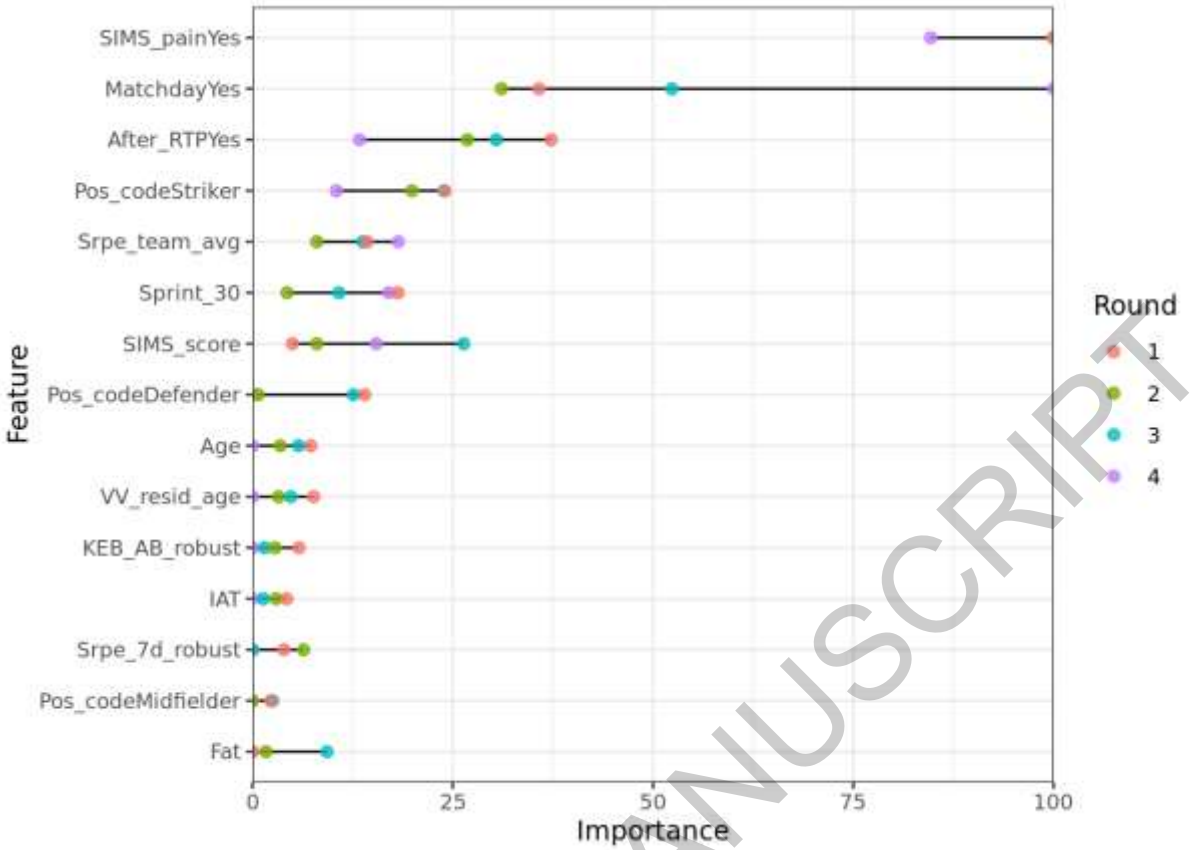

Figure 7 – Feature importance across round-robin repetitions

x-axis scaling: 100 indicates most important feature in the particular model

<u>Comparator models</u>

Model evaluation results of the comparator models with monitoring data only and monitoring data plus basic player information (table 3) are inferior to the performance of the full model and thereby in support of integrating screening and monitoring data. This applies particularly for test set performance.

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Table 3: Comparator models		
	ROC AUC	
	Cross-validation	Test set performance
Monitoring features only	0.580 [0.570 | 0.590]	0.558 [0.385 | 0.732]
Monitoring features plus basic player information	0.583 [0.573 | 0.593]	0.516 [0.337 | 0.695]
ROC AUC: Estimate [95% confidence interval (bootstrap)] ROC: receiver operating characteristic curve; AUC: area under the curve Only difference to main analysis: subsetting of explanatory features		

Discussion

In this work we describe the development and evaluation of a forecasting algorithm for acute, non-contact, time-loss injuries in professional football which aims to leverage the "web of determinants" concept by integrating screening and monitoring parameters. According to common standards (69-71), the area under the ROC curve (ROC-AUC) for the gradient boosted model (table 2) indicates fair performance in forecasting injuries. Of note, the small difference in ROC-AUC from cross-validation to the holdout test set is in support of effective prevention of information leakage and potential generalizability.

From a comparative standpoint, performance is superior to the within-study comparator models without the screening features (table 3). This points to added value from integrating (periodic) screening and (daily) monitoring data. Like the within-study comparator models, previous published attempts on machine learning based injury forecasting in professional adult football have been limited to monitoring data - with (15, 47) or without (48) basic player characteristics. Reported cross-validation ROC-AUC ranges between 0.75 and 0.80 (15, 47, 48) and is are thereby higher than cross-validation results for our main model. However, the comparison is complicated by a lack of reported test set results and reproducible details of model fitting (e.g. in shared code). Beyond feature selection and hyperparameter tuning, this point specifically concerns the prevention of information leakage between cross-validation folds. Ancillary analysis 2 (table 2) demonstrates the large and spurious gain in cross-validation ROC-AUC when upsampling is performed globally before model training and folds are generated on the level of datapoints (ignoring the nesting of timepoints within players). In (47) the pitfall of information leakage between cross-validation folds is discussed. However, it does not become unambiguously clear wether cross-validation folds are generated on the level of datapoints or players. In other cases, (15, 48) the issue is not explicitly addressed and the hierarchical data structure does not seem to be taken into

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account (fig. 1 in (15)). Similar considerations apply for injury forecasting in youth football (72, 73). However, direct comparability of predictive performance is further limited e.g. by the role of biological maturation and related features. Taken together, improving transparency with respect to technical details as well as full model evaluation is needed to enable meaningful comparison between approaches. This is particularly important in light of the complicated framework conditions for machine learning based injury forecasting in football (high-dimensional causal concepts, small n, hierarchical data structure, marked imbalanced ("needle in a haystack prediction")) and includes proactively probing expectable weak points (e.g. robustness of results).

From the perspective of potential application in sports practice, an intuitive indicator of model performance is the proportion of training bouts that would have to be altered in order to (presumably) prevent a relevant share of criterion injuries ("proportion needed to prevent"). As illustrated in figure 5, the ratio between identified (and thereby potentially prevented) injuries and flagged time-points is ~ 4 for predictions on the full dataset. For pooled test set predictions the ratio is only only ~ 1.5, however pooling predictions based on different models (with likely different calibration) likely leads to an underestimation. While agreed thesholds for interpretation are lacking (and would have to be specific for the level of play and framework conditions (43)), we argue that this could be a meaningful contribution to informing decision making in sports practice. By contrast, fixed consequences based on the predicted outcome alone (e.g. exclusion from matchplay above a specific probability prediction of injury occurence) are clearly not warranted.

When considering machine learning based injury forecasting as part of the groundwork for informed human decision making, this calls for probability predictions, thereby avoiding the (in this case unnecessary) loss of information associated with a dichotomous classification. Importantly, the requirements for probability predictions meant to support decision making depend on the type of decisions to be made and their field of application (74, 75). Therefore, it is important to keep in mind that the framework conditions in high-level football differ considerably from clinical settings. For example, for each competitive match 11 players will have to be lined up – even if for all members of the squad the predicted injury probability is high (e.g. due to match congestion). Likewise, even a player with a high individual level of injury risk (e.g. due to many previous injuries) will have to be exposed to physical load – or quit the job. By contrast, in clinical settings decisions are made independently for each patient with the consequences for his / her health as the sole criterion. Consequently, it may be argued that in sports practice decisions will likely be influenced by a relative interpretation of the probability predictions between players and / or time-points. This requires that the model assigns higher risk-estimates to player-days with injury occurrence ("discrimination",

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figure 4). By contrast clinical decisions are typically based on the absolute value of the probability estimate (the individual´s “risk” of having or developing e.g. cancer (76)).

Beyond discrimination, such decisions also require calibration (76, 77) – the numerical agreement between the estimated probability of an event and its actually observed relative frequency. These applied considerations have important consequences for the development of the machine learning algorithm. While our model shows fair discrimination (figure 4, figure 3, table 2), calibration is poor with an overestimation of injury risk (cp. calibration plots in the .Rmd file available at doi: https://zenodo.org/badge/latestdoi/493978425). This may be expected based on the relatively small sample size and the marked underrepresentation of the minority class (“needle in a haystack”). We use upsampling of the training sets to mitigate the negative effects of imbalance on model training – namely to improve discrimination in the test set as measured by ROC-AUC. As a "side effect", the model also “learns” the artificially increased overall proportion of days with injury (68) – aggravating the overestimation of injury probability. It has been argued recently that upsampling does not only cause (or aggravate) miscalibration but is also ineffective in improving discrimination (and should therefore be abandoned) (68). However, we hold that in this specific case upsampling is a meaningful aid for two main reasons: (i) Cross-validation Test set ROC-AUC is markedly improved when upsampling is used (table 2 and ancillary analysis 3). Potentially, the discrepancy to the results of the above mentioned simulation study (68) may be due to logistic regression beeing less prone to overfitting as compared to gradient boosting. (ii) As explained above, the relative importance of discrimination and calibration differs between football and the clinical setting. Moreover linear rescaling of probability predictions using the proportion of days with injury in the non-upsampled training set could be used to revert the model learning an artificially increased proportion of days with injury (cp. calibration plots in the knit Markdown file for the main analysis).

Lastly, model performance may be interpreted on the background of what may be expected from a best-possible algorithm taking into account the suspected degree of aleatoric (irreducible) uncertainty as well as real-world epistemical constraints (number of, uncertainty in, and correlations between explanatory features; small case numbers etc). Arguably, aleatoric uncertainty of injury occurrence is considerable (“Shit happens!”) not only for contact but also for non-contact injuries. Consequently, even a best-possible algorithm will not be able to prospectively forecast injury occurrence in football with near perfect accuracy. This has to be kept in mind when assessing claims made for commercial monitoring and injury risk mitigation systems. The lack in transparent documentation of proprietary algorithms compounds the importance of this consideration (78, 79).

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While even the pooled test set results indicate fair model performance and forecasting accuracy, the large variability in performance metrics and feature importance across the round-robin repetitions of the model fitting procedure point to a lack of robustness, presumably due to the relatively low number of criterion injuries. As a consequence, prospective verification and continued model development with accumulating numbers of injuries are mandatory before a transfer to productive operation in sports practice may be envisaged. A “pseudo-prospective” temporal validation has not been attempted, because the number of features has been based on the expected number of injuries during a full season. Importantly, the present study aims at a proof-of-concept and not at a final model with stable model parameters ready for productive operation (which would be foolish with the available number of cases).

Regarding the within-study (table 3) as well as published comparator models (15, 47), it is important to bear in mind that these are all based on monitoring data (with or without consideration of basic player characteristics) but do not include screening results. Therefore, the performance of our main model is in support of integrating screening and monitoring data. While in our analysis the monitoring features seems to gradually modify the baseline risk set by screening features and exposure characteristics (cp. figures 6 and 7) it can of couse not be ruled out that similar or superior predictive performance is possible based on monitoring data only. Of note, nonlinear models based exclusively on screening data (e.g. (44-46, 80) estimate injury risk for a prolonged period of time (and not on a daily basis). Direct comparison to our work is therefore not possible. Indirect support for the integration of screening and monitoring data comes from the time courses of probability predictions (figure 6) which show the expected volatile variations around an individual level of injury risk set by time invariant features. Regarding the potential for application in sports practice, the considerable effort associated with collecting screening as well as monitoring data has to be considered. However, at least in professional football, screening as well as monitoring data are already routinely collected. Integrating these so far separate data sources may provide the basis for our (or similar) algorithms without causing additional testing burden for the players or staff. Of couse, the algorithm (including feature engineering and modelling) has to be adapted to the tests and scales used in a specific club or team. Moreover, the consequences of changes in the details of data collection for forecasting accuracy remain to be evaluated.

The criterion for model performance in this work is forecasting accuracy. A competing criterion is interpretability (that is: getting to know which feature drives a specific estimate of injury risk). While interpretability would be highly desirable in particular from the practitioners

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perspective, making trade-offs between accuracy and interpretability may lead to impaired performance regarding either criterion. Results for plausible alternative approaches to data analysis are provided in ancillary analysis 4 (cp repository cited above). Of note: with the current dataset logistic regression offers comparable (numerically superior) model performance. While the law of parsimony would suggest to use logistic regression, its current lead may mirror the higher variance of gradient boosting (which is therefore likely to profit more from accumulating data over prospective seasons). However, this remains to be verified. The relative importance of features in the trained models (as well as the variation with different data splits) is visualized in figure 7. When comparing feature importance to expectations from univariate analyses, it has to be kept in mind that correlated features compete for a common share of importance. An example are the two exposure features (team-average of sessionRPE and match vs. training). Replicability of the the lower-than-expected importance of injury history and monitoring features remains to be verified in follow-up studies.

Regarding injury incidence, a recent systematic review reports a range of 2.5 - 9.4 per 1000 hours of exposure for total injury incidence in male, adult professional football (competitive matches: 8.7 – 65.9; training: 1.4 – 5.8). For all three categories (overall, match, training), the incidence of time-loss injuries observed in the present study is near the lower end of that broad spectrum. As most of the respective research has been done in elite leagues, a difference in framework conditions may play a role. This concerns primarily potential influencing factors of injury risk (e.g. match congestion, international tournaments, long-distance travel etc.), but differences in the consistency of reporting for (minor) time-loss injuries also seem conceivable. Lastly, random season-by-season variation may play a role.

Conclusion

Model performance indicates the potential for a meaningful contribution of this approach to decision making in sports practice. Plausibly, the added value is based on the integration of explanatory features from all categories. Beyond stable screening and volatile monitoring features this includes basic player and exposure characteristics (figure 1). This integrated dataset is leveraged by gradient boosting tuned to accommodate the specific constraints of injury forecasting in football. Importantly, verification and further development is required before application in sports practice may be envisaged.

Acknowledgements

The authors wish to thank Alexander Ferrauti, Mark Pfeiffer, and Michael Kellmann for their support of data collection.

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Funding

The present study was funded by the German Institute of Sport Science within the project "REGman - Management of Regeneration in Elite Sports" (IIA1-081901/17-20).

Disclosure of interest

The authors report no conflict of interest

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